RESEARCH IN SCIENCE & TECHNOLOGICAL EDUCATION

01/03

NEW ARTICLE

Enhancing science learning through choice and prior knowledge: a mixed-methods study on fluvial processes using a stream table

Shing-Fung Chung[a] and Chi-Chang Liu[b]

[a] WWF Hong Kong, China

[b] National Taiwan University, Taiwan

DOI **doi.org/10.1080/02635143.2026.2755470**

Taylor & Francis
Taylor & Francis Group

WHAT DID THE RESEARCHERS INVESTIGATE?

- Learning in science may depend on students' individual background and the autonomy the learning environment gives them.
- Using Falk and Dierking's Contextual Model of Learning, the authors examined prior knowledge and degrees of choice control during a hands-on activity.
- Thirty-six university students ran simulated experiments on fluvial processes with a stream table.

Chung and Liu (2026) · Research in Science & Technological Education · doi.org/10.1080/02635143.2026.2755470

WHAT DID THEY FIND?

- Both groups showed significant gains in interest and knowledge; students with prior lessons used correct scientific terminology more often.
- A limited choice setting fostered the most collaborative problem-solving, compared with no choice.
- Free choice stimulated the greatest curiosity and creativity but often led to off-topic engagement.

Chung and Liu (2026) · Research in Science & Technological Education · doi.org/10.1080/02635143.2026.2755470